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SARDAR PATEL UNIVERSITY

External Examination NC - April 2025

M.Sc. (Information Technology) Semester - I NC

PS01CINT51 - Object Oriented Programming using JAVA

Time: 10:00 am to 1:00 pm

Date: 08/04/2025

Marks: 70

Q.1 Choose the most appropriate option for each questions.

[8]

1. What is full form of JVM?
[A]. Java View Model [B]. Java Virtual Memory
[C]. Java Variable Model [D]. Java Virtual Machine
2. Java compiler produce intermediate code known as _____.
[A]. Bit code [B]. Machine code
[C]. Byte code [D]. User code
3. Which method is used to start the execution of a Java program?
[A]. run() [B]. main()
[C]. execute() [D]. start()
4. In Java, which keyword is used to inherit a class?
[A]. extends [B]. implement
[C]. inherits [D]. uses
5. Which of the following Swing components is used to display a single line of text that the user can edit?
[A]. JTextArea [B]. JLabel
[C]. JTextField [D]. JButton
6. Which layout manager in Swing arranges components at the top, bottom, left, right, and center of the container?
[A]. GridLayout [B]. CardLayout
[C]. FlowLayout [D]. BorderLayout
7. Which of the following is used to establish a connection to the database in JDBC?
[A]. getConnection() [B]. createStatement()
[C]. executeQuery() [D]. setAutoCommit()
8. What is the main advantage of using a Type-4 JDBC driver?
[A]. It requires native database client software. [B]. It translates Java calls directly into database-specific protocol calls.
[C]. It uses a two-tier architecture for database connectivity. [D]. It is the slowest among all JDBC driver types.

Q2. Answer the following questions.(any seven)

1. What is Auto-boxing and Unboxing in Java? Explain with examples.
2. Explain the types of comments in JAVA.
3. Define inheritance in Java. What are the different types of inheritance supported?
4. Write a short note on this and super keyword
5. What is an error in Java? How is it different from an exception?
6. Differentiate between AWT and Swing components in Java.
7. List commonly used Swing components in Java.
8. What is JDBC in Java? Explain its purpose and list the types of JDBC drivers.
9. Explain the ResultSet object in JDBC. How is it used to retrieve data from a database?

Q3. Answer the following questions.

[6]

[A] Explain decision operators in JAVA.

[6]

[B] Explain JAVA features.

Or

[6]

[B] Write Looping Statement with syntax and example.

Q4. Answer the following questions.

[6]

[A] What is Exception? Explain try and catch block.

[6]

[B] Explain Multithreading.

Or

[6]

[B] Explain File Handling.

Q5. Answer the following questions.

[6]

[A] Define Swing components. Explain JButton.

[6]

[B] Write a short note on jCheckBox and JRadioButton

Or

[6]

[B] What is Layout in Swing? Explain Border Layout.

Q6. Answer the following questions(any seven):

[6]

[A] What is JDBC? Explain JDBC Drivers.

[6]

[B] What is JDBC? Explain the architecture of JDBC.

Or

[6]

[B] Explain steps to achieve Java Database Connectivity

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